

Lecture 6: Surface elements and the first fundamental form

We now pass from curves to surfaces.

For curves, regularity meant that the derivative never vanished. For surfaces, the corresponding condition is that two independent tangent directions survive under the derivative.

Definition. Let $U \subset \mathbb{R}^2$ be open. A smooth map

$$f : U \longrightarrow \mathbb{R}^3$$

is an ***immersion*** if its derivative

$$Df_u : \mathbb{R}^2 \longrightarrow \mathbb{R}^3$$

is injective for every $u \in U$.

Equivalently,

$$\text{rank } Df_u = 2.$$

An immersion may have self-intersections. The condition concerns the derivative, not the global behavior of the map.

Write the coordinates on U as (u, v) , and write

$$f(u, v) = (x(u, v), y(u, v), z(u, v)).$$

The columns of the derivative are

$$f_u = \frac{\partial f}{\partial u}, \quad f_v = \frac{\partial f}{\partial v}.$$

Thus f is an immersion precisely when

$$f_u(u, v), f_v(u, v)$$

are linearly independent at every point. In $\mathbb{R}^3$, this is equivalent to

$$f_u \times f_v \neq 0.$$

A point where this condition fails is called a ***singular point*** of the parametrization.

Example. Let $h : U \rightarrow \mathbb{R}$ be smooth. Its graph is parametrized by

$$f(u, v) = (u, v, h(u, v)).$$

Then

$$f_u = (1, 0, h_u), \quad f_v = (0, 1, h_v),$$

and

$$f_u \times f_v = (-h_u, -h_v, 1).$$

The last component is always 1, so the cross product never vanishes. Every graph is therefore regularly parametrized.

The graph example is special: projection onto the first two coordinates recovers the parameters. General parametrized surfaces need not be graphs globally.

Definition. A *parametrized surface element* in $\mathbb{R}^3$ is an immersion

$$f : U \longrightarrow \mathbb{R}^3, \quad U \subset \mathbb{R}^2 \text{ open.}$$

The map f is also called a parametrization.

The points $(u, v) \in U$ are the parameters, while $f(u, v)$ is the corresponding point in space.

Fixing one parameter gives the coordinate curves

$$u \longmapsto f(u, v_0), \quad v \longmapsto f(u_0, v).$$

Their velocity vectors at (u_0, v_0) are f_u and f_v .

Thus a coordinate grid in U is carried to a curvilinear grid on the surface element.

A parametrization contains more information than the geometric surface element: it includes a choice of coordinates.

Definition. Two parametrized surface elements

$$f : U \rightarrow \mathbb{R}^3, \quad \tilde{f} : \tilde{U} \rightarrow \mathbb{R}^3$$

are equivalent if there is a diffeomorphism

$$\phi : \tilde{U} \longrightarrow U$$

such that

$$\tilde{f} = f \circ \phi.$$

A ***non-parametrized surface element*** is an equivalence class of parametrized surface elements.

The map ϕ changes the coordinates, but not the geometric points described by the surface element.

Example. The paraboloid $z = x^2 + y^2$ can be described using Cartesian or polar coordinates. On its portion with $x > 0$, set

$$f(u, v) = (u, v, u^2 + v^2), \quad (u, v) \in U = (0, \infty) \times \mathbb{R},$$

$$\tilde{f}(r, \theta) = (r \cos \theta, r \sin \theta, r^2), \quad (r, \theta) \in V = (0, \infty) \times (-\pi/2, \pi/2).$$

The change of coordinates is

$$\phi(r, \theta) = (r \cos \theta, r \sin \theta), \quad \tilde{f} = f \circ \phi.$$

It is a diffeomorphism $V \rightarrow U$, with smooth inverse

$$\phi^{-1}(u, v) = \left(\sqrt{u^2 + v^2}, \arctan(v/u) \right).$$

Indeed,

$$D\phi = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}, \quad \det D\phi = r > 0.$$

Thus these are equivalent surface elements. The curves with fixed r are horizontal circular arcs. Those with fixed θ are arcs of vertical parabolas.

Why exclude $r = 0$? There $\tilde{f}_\theta = 0$, so the polar map is not an immersion. The full paraboloid is nevertheless smooth at the origin: its Cartesian graph parametrization remains regular.

Let $f : U \rightarrow \mathbb{R}^3$ be a parametrized surface element and let $u \in U$.

Definition. The *tangent space* of f at u is

$$T_u f = \text{im } Df_u \subset \mathbb{R}^3.$$

Since Df_u has rank 2, this is a two-dimensional vector subspace. In coordinates,

$$T_u f = \text{span}\{f_u(u), f_v(u)\}.$$

The corresponding affine tangent plane through $p = f(u)$ is

$$p + T_u f.$$

If the image is an embedded surface S near p , we write this space as $T_p S$.

Definition. A *tangent vector* to the surface element at u is a vector in $T_u f$.

If $\xi = (a, b) \in \mathbb{R}^2$, then

$$Df_u(\xi) = af_u(u) + bf_v(u).$$

Thus (a, b) are the coordinates of the tangent vector relative to the basis (f_u, f_v) .

Tangent vectors are also velocities of curves on the surface. If

$$\gamma : (-\varepsilon, \varepsilon) \rightarrow U, \quad \gamma(0) = u,$$

and $c = f \circ \gamma$, then

$$\dot{c}(0) = Df_u(\dot{\gamma}(0)) \in T_u f.$$

Conversely, every vector in $T_u f$ arises in this way.

Definition. The *normal space* at u is

$$N_u f = (T_u f)^\perp.$$

It is one-dimensional. Any nonzero vector in $N_u f$ is a normal vector.

The two unit normal vectors are

$$\nu(u) = \frac{f_u \times f_v}{\|f_u \times f_v\|} \quad \text{and} \quad -\nu(u).$$

Choosing one of these signs smoothly is called choosing an orientation of the surface element.

The tangent and normal spaces give an orthogonal decomposition

$$\mathbb{R}^3 = T_u f \oplus N_u f.$$

Suppose $\tilde{f} = f \circ \phi$, where

$$\phi(r, s) = (u(r, s), v(r, s)).$$

The chain rule gives

$$\begin{aligned}\tilde{f}_r &= f_u u_r + f_v v_r, \\ \tilde{f}_s &= f_u u_s + f_v v_s.\end{aligned}$$

Since $D\phi$ is invertible, the two pairs

$$(f_u, f_v) \quad \text{and} \quad (\tilde{f}_r, \tilde{f}_s)$$

span the same tangent space.

Moreover,

$$\tilde{f}_r \times \tilde{f}_s = \det(D\phi) (f_u \times f_v) \circ \phi.$$

Thus the normal line is unchanged. The preferred unit normal changes sign precisely when ϕ reverses orientation.

A parametrized surface element need not be injective. Its image may cross itself, and two parameter points mapping to the same point of $\mathbb{R}^3$ may represent different tangent planes.

Nevertheless, every immersion is locally an embedding.

Local form of an immersion. If $f : U \rightarrow \mathbb{R}^3$ is an immersion and $u_0 \in U$, then there is a neighborhood $V \subset U$ of u_0 such that

$$f|_V : V \longrightarrow \mathbb{R}^3$$

is an embedding onto its image.

Idea of the proof. Two components of f have an invertible 2×2 Jacobian at u_0 . The inverse function theorem makes those two components into local coordinates. In these coordinates, the remaining component is a graph.

Definition. A subset $S \subset \mathbb{R}^3$ is a smooth **surface**, or an embedded two-dimensional submanifold, if every point $p \in S$ has a neighborhood in which S is the image of an embedding

$$f : U \subset \mathbb{R}^2 \longrightarrow \mathbb{R}^3.$$

Equivalently, for each $p \in S$ there is an open neighborhood $W \subset \mathbb{R}^3$ and a smooth function $F : W \rightarrow \mathbb{R}$ such that

$$S \cap W = F^{-1}(0), \quad \nabla F \neq 0 \text{ on } F^{-1}(0).$$

Thus:

- a parametrized surface element is a local description;
- an immersion may have global self-intersections;
- a submanifold is locally embedded and is covered by compatible parametrizations.

For example, the sphere

$$S_R^2 = \{(x, y, z) : x^2 + y^2 + z^2 = R^2\}$$

is a submanifold. Set

$$F(x, y, z) = x^2 + y^2 + z^2 - R^2.$$

Then

$$\nabla F(x, y, z) = 2(x, y, z),$$

which never vanishes on S_R^2 .

At $p \in S_R^2$, a tangent vector X satisfies

$$dF_p(X) = 2\langle p, X \rangle = 0.$$

Therefore

$$T_p S_R^2 = p^\perp,$$

and the radial vector p is normal to the sphere.

This description agrees with the tangent space obtained from any local parametrization.

Examples. The cylinder of radius $R > 0$,

$$S = \{(x, y, z) : x^2 + y^2 = R^2\},$$

is an embedded surface. For $H(x, y, z) = x^2 + y^2 - R^2$,

$$\nabla H = (2x, 2y, 0) \neq 0 \quad \text{on } S.$$

At $p = (x, y, z) \in S$,

$$T_p S = \{(a, b, c) : xa + yb = 0\}.$$

The radial vector $(x, y, 0)$ is normal; the vertical direction $(0, 0, 1)$ is tangent everywhere.

The saddle $S = \{z = xy\}$ is also embedded. Here

$$H(x, y, z) = z - xy, \quad \nabla H = (-y, -x, 1) \neq 0.$$

At $p = (u, v, uv)$, its tangent space is

$$T_p S = \{(a, b, c) : c = va + ub\}.$$

Its graph parametrization $f(u, v) = (u, v, uv)$ is a global embedding: projection to the first two coordinates is its inverse. We will compute both surfaces' first fundamental forms below.

The Euclidean inner product in $\mathbb{R}^3$ measures lengths and angles of tangent vectors. To an embedded surface S we obtain a natural inner product on $T_p S$ for every $p \in S$.

Definition. The *first fundamental form* of a parametrized surface element $f : U \rightarrow \mathbb{R}^3$ is

$$I_u(\xi, \eta) = \langle Df_u(\xi), Df_u(\eta) \rangle, \quad \xi, \eta \in \mathbb{R}^2.$$

It is the pullback of the Euclidean inner product by f .

Equivalently, on the tangent space itself,

$$I_p(X, Y) = \langle X, Y \rangle_{\mathbb{R}^3}, \quad X, Y \in T_p S.$$

Because Df_u is injective, I_u is positive definite. It is therefore an inner product on each tangent space.

Relative to the coordinate basis, define

$$E = \langle f_u, f_u \rangle, \quad F = \langle f_u, f_v \rangle, \quad G = \langle f_v, f_v \rangle.$$

The matrix of the first fundamental form is

$$g = \begin{pmatrix} E & F \\ F & G \end{pmatrix} = (Df)^\top Df.$$

For $\xi = (a, b)$,

$$I_u(\xi, \xi) = Ea^2 + 2Fab + Gb^2.$$

One often writes the same quadratic form symbolically as

$$ds^2 = E du^2 + 2F du dv + G dv^2.$$

The symbols du, dv record components of tangent vectors; they are not small finite displacements.

The immersion condition appears directly in the first fundamental form. By the cross-product identity,

$$\|f_u \times f_v\|^2 = \|f_u\|^2 \|f_v\|^2 - \langle f_u, f_v \rangle^2.$$

Thus

$$\det g = EG - F^2 = \|f_u \times f_v\|^2 > 0.$$

Consequently,

$$E > 0, \quad G > 0, \quad EG - F^2 > 0.$$

The area of the parallelogram spanned by f_u and f_v is

$$\sqrt{EG - F^2}.$$

This leads to the surface-area element

$$dA = \sqrt{EG - F^2} \, du \, dv.$$

Let $\gamma(t) = (u(t), v(t))$ be a curve in the parameter domain and let

$$c(t) = f(\gamma(t))$$

be the corresponding curve on the surface. Then

$$\dot{c} = f_u \dot{u} + f_v \dot{v}.$$

Therefore

$$\|\dot{c}(t)\|^2 = E\dot{u}^2 + 2F\dot{u}\dot{v} + G\dot{v}^2 = I_{\gamma(t)}(\dot{\gamma}, \dot{\gamma}).$$

The length of c is consequently

$$L(c) = \int_a^b \sqrt{E\dot{u}^2 + 2F\dot{u}\dot{v} + G\dot{v}^2} dt.$$

Thus the first fundamental form contains exactly the information needed to measure lengths of curves on the surface.

The first fundamental form also measures angles. For nonzero tangent vectors X, Y ,

$$\cos \theta = \frac{I(X, Y)}{\sqrt{I(X, X)}\sqrt{I(Y, Y)}}.$$

In particular, the coordinate curves meet orthogonally precisely when

$$F = \langle f_u, f_v \rangle = 0.$$

Remarks.

- The first fundamental form depends only on first derivatives of the parametrization.
- It records lengths, angles, and areas within the surface.
- It is not generally the standard inner product on the parameter domain U .
- Its coefficients E, F, G depend on the chosen coordinates, although the geometry they describe does not.

For the graph

$$f(u, v) = (u, v, h(u, v)),$$

we found

$$f_u = (1, 0, h_u), \quad f_v = (0, 1, h_v).$$

Hence

$$E = 1 + h_u^2, \quad F = h_u h_v, \quad G = 1 + h_v^2.$$

The first fundamental form is

$$ds^2 = (1 + h_u^2) du^2 + 2h_u h_v du dv + (1 + h_v^2) dv^2.$$

Even though the parameter domain is flat, the graph generally stretches lengths in a way that depends on the derivatives of h .

Example: the saddle. For the embedded graph $f(u, v) = (u, v, uv)$,

$$f_u = (1, 0, v), \quad f_v = (0, 1, u).$$

Taking the three inner products gives

$$E = 1 + v^2, \quad F = uv, \quad G = 1 + u^2.$$

Thus

$$ds^2 = (1 + v^2) du^2 + 2uv du dv + (1 + u^2) dv^2,$$

and

$$EG - F^2 = 1 + u^2 + v^2 > 0.$$

For instance, at $f(1, 1) = (1, 1, 1)$, the coordinate tangent vectors are $(1, 0, 1)$ and $(0, 1, 1)$. Each has length $\sqrt{2}$, and

$$\cos \alpha = \frac{F}{\sqrt{EG}} = \frac{1}{2}.$$

They meet at angle $\pi/3$, even though the parameter axes meet at a right angle.

The tangent vector with coordinate components $(1, -1)$ is $f_u - f_v = (1, -1, 0)$. Its squared length is

$$E - 2F + G = 2,$$

agreeing with its Euclidean squared length in $\mathbb{R}^3$.

Consider the unit sphere parametrized by longitude and latitude:

$$f(\varphi, \vartheta) = (\cos \varphi \cos \vartheta, \sin \varphi \cos \vartheta, \sin \vartheta).$$

Take $-\pi < \varphi < \pi$ and $-\pi/2 < \vartheta < \pi/2$ for an embedded coordinate patch. Differentiating gives

$$\begin{aligned} f_\varphi &= (-\sin \varphi \cos \vartheta, \cos \varphi \cos \vartheta, 0), \\ f_\vartheta &= (-\cos \varphi \sin \vartheta, -\sin \varphi \sin \vartheta, \cos \vartheta). \end{aligned}$$

Their inner products are

$$E = \cos^2 \vartheta, \quad F = 0, \quad G = 1.$$

Thus

$$ds^2 = \cos^2 \vartheta d\varphi^2 + d\vartheta^2.$$

A change $d\varphi$ in longitude has physical length $\cos \vartheta |d\varphi|$. Circles of latitude shrink toward the poles.

At $\vartheta = \pm\pi/2$, the coefficient E vanishes and the parametrization is not an immersion. The sphere is smooth there; this particular coordinate system is singular.

Example: the cylinder. For the cylinder $x^2 + y^2 = R^2$, use the embedded patch

$$f(\theta, z) = (R \cos \theta, R \sin \theta, z), \quad -\pi < \theta < \pi, \quad z \in \mathbb{R}.$$

This patch omits one vertical line. Other angular intervals give patches covering that line as well. The coordinate tangent vectors are

$$f_\theta = (-R \sin \theta, R \cos \theta, 0), \quad f_z = (0, 0, 1).$$

Consequently,

$$E = R^2, \quad F = 0, \quad G = 1, \quad ds^2 = R^2 d\theta^2 + dz^2.$$

Putting $\ell = R\theta$ gives $ds^2 = d\ell^2 + dz^2$, the Euclidean metric on a strip. Thus unrolling a cylinder preserves lengths and angles locally, although the cylinder bends in $\mathbb{R}^3$.

For a concrete length calculation, consider the helix

$$c(t) = (R \cos t, R \sin t, bt).$$

Its parameter velocity is $(\dot{\theta}, \dot{z}) = (1, b)$, so

$$\|\dot{c}(t)\|^2 = R^2 + b^2, \quad L(c|_{[0, 2\pi]}) = 2\pi\sqrt{R^2 + b^2}.$$

One full turn passes through more than one coordinate patch; the same local formula computes its length throughout.

Example: a cone without its vertex. Fix $a > 0$. The upper cone

$$S = \{(x, y, z) : z = a\sqrt{x^2 + y^2}, z > 0\}$$

is an embedded surface: it is a smooth graph over $\mathbb{R}^2 \setminus \{0\}$. A polar coordinate patch is

$$f(r, \theta) = (r \cos \theta, r \sin \theta, ar), \quad r > 0, \quad -\pi < \theta < \pi.$$

Since

$$f_r = (\cos \theta, \sin \theta, a), \quad f_\theta = (-r \sin \theta, r \cos \theta, 0),$$

we get

$$E = 1 + a^2, \quad F = 0, \quad G = r^2,$$

and therefore

$$ds^2 = (1 + a^2) dr^2 + r^2 d\theta^2, \quad dA = r\sqrt{1 + a^2} dr d\theta.$$

A radial segment from r_0 to $r_1 > r_0$ has length $\sqrt{1 + a^2}(r_1 - r_0)$; a horizontal circle has circumference $2\pi r$.

Adding the vertex does not give a smooth surface there. The ray directions $(\cos \theta, \sin \theta, a)$ cannot all lie in one two-dimensional tangent plane at the vertex.

We now verify that the first fundamental form is independent of the chosen parametrization.

Lemma. Let

$$\tilde{f} = f \circ \phi, \quad \phi : \tilde{U} \longrightarrow U$$

be a change of coordinates. If g and $\tilde{g}$ are the matrices of the first fundamental forms of f and $\tilde{f}$, then

$$\boxed{\tilde{g}(q) = D\phi(q)^{\top} g(\phi(q)) D\phi(q).}$$

Equivalently, for $\xi, \eta \in \mathbb{R}^2$,

$$\tilde{I}_q(\xi, \eta) = I_{\phi(q)}(D\phi_q \xi, D\phi_q \eta).$$

The coefficients change, but the inner product on geometric tangent vectors does not.

Proof. The chain rule gives

$$D\tilde{f}_q = Df_{\phi(q)}D\phi_q.$$

Therefore

$$\begin{aligned}\tilde{\mathbf{I}}_q(\xi, \eta) &= \langle D\tilde{f}_q\xi, D\tilde{f}_q\eta \rangle \\ &= \langle Df_{\phi(q)}D\phi_q\xi, Df_{\phi(q)}D\phi_q\eta \rangle \\ &= \mathbf{I}_{\phi(q)}(D\phi_q\xi, D\phi_q\eta).\end{aligned}$$

In matrix notation,

$$\begin{aligned}\tilde{g} &= (D\tilde{f})^\top D\tilde{f} \\ &= (D\phi)^\top (Df)^\top Df (D\phi) \\ &= (D\phi)^\top g(D\phi).\end{aligned}$$

This proves the lemma.

Example: the paraboloid in two coordinate systems. Return to the paraboloid from the earlier example,

$$f(u, v) = (u, v, u^2 + v^2).$$

The graph formula gives

$$g(u, v) = \begin{pmatrix} 1 + 4u^2 & 4uv \\ 4uv & 1 + 4v^2 \end{pmatrix}.$$

On the portion $u > 0$, introduce polar coordinates:

$$\tilde{f}(r, \theta) = (r \cos \theta, r \sin \theta, r^2), \quad r > 0, \quad -\pi/2 < \theta < \pi/2.$$

Direct differentiation gives

$$\tilde{f}_r = (\cos \theta, \sin \theta, 2r), \quad \tilde{f}_\theta = (-r \sin \theta, r \cos \theta, 0).$$

Thus

$$\tilde{E} = 1 + 4r^2, \quad \tilde{F} = 0, \quad \tilde{G} = r^2,$$

and

$$ds^2 = (1 + 4r^2) dr^2 + r^2 d\theta^2.$$

The polar coordinate curves are orthogonal. In Cartesian coordinates, the cross term $F = 4uv$ is usually nonzero. These are different coordinate grids on the same surface.

Verifying the coordinate-change formula.

For $\phi(r, \theta) = (r \cos \theta, r \sin \theta)$, set

$$A = D\phi = \begin{pmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{pmatrix}, \quad w = \begin{pmatrix} r \cos \theta \\ r \sin \theta \end{pmatrix}.$$

The Cartesian metric, evaluated at $\phi(r, \theta)$, is

$$g \circ \phi = I_2 + 4ww^\top.$$

The columns of A are orthogonal, and direct multiplication gives

$$A^\top A = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}, \quad A^\top w = \begin{pmatrix} r \\ 0 \end{pmatrix}.$$

Hence the lemma predicts

$$\begin{aligned} \widetilde{g} &= A^\top (I_2 + 4ww^\top) A \\ &= \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix} + 4 \begin{pmatrix} r^2 & 0 \\ 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} 1 + 4r^2 & 0 \\ 0 & r^2 \end{pmatrix}, \end{aligned}$$

exactly as in the direct computation.

In particular, a vector with polar components (a, b) has squared length $(1 + 4r^2)a^2 + r^2b^2$. Its Cartesian components are $A(a, b)^\top$; using g gives the same squared length.